PROJECT 5: Clinic SOP Copilot with MCP

Tool-based Agent via Model Context Protocol

Scenario
Clinics need SOP-consistent answers (e.g., "How do we prep a patient for MRI with contrast?"). Build an MCP server exposing:

- a *filesystem tool* (reads sops/*.md)
- a *policy QA tool* (structured search over SOP headings)
- a *web tool* (Serper search for reputable references)

Primary stack: MCP (server + client), your LLM of choice, Opik for eval
MCP Tools

- sop.search(query) → returns section IDs & excerpts
- sop.read(section_id) → returns full section
- web.search(query) → Serper JSON results (title/url/snippet)
- cite.validate(text, citations) → simple checker to ensure each claim has a source

Client

- Minimal CLI or VS Code or Claude Desktop integrated sample that calls MCP tools before responding

Deliverables

- mcp-server/ (Node/TS or Python)
- client/ with a "force-tool-use" system prompt and conversation log
- tests/ with probe questions → expected section IDs & allowed answers

Evaluation (Opik)

- *Tool Compliance*: % turns that used a tool before answering
- *Citation Coverage*: % sentences with valid citation anchors
- *Answer Consistency*: exact-match scores vs. canonical SOP snippets

Team of 4 split
1 MCP server/tools, 1 client & UX, 1 SOP content & indexing, 1 eval/Opik hooks

Stretch

- Add guardrails: block answers if no SOP match ≥ threshold
- Add "uncertainty mode" that asks the user to clarify

SOPs dataset

Synthetic SOPs are provided

How to use with MCP tools

- **sop.search(query)**: search titles, keywords, and [SECTION: …] blocks. Return section IDs like SOP-001::PROCEDURE.
- **sop.read(section_id)**: map SOP-XXX::SECTION_NAME → full section text.
- **cite.validate(text, citations)**: ensure answers include anchors such as SOP-001 [SECTION: PROCEDURE].